

Energy Consumption Analysis using MySQL & Power BI

❖ Objective:

- To build an end-to-end data pipeline integrating MySQL and Power BI for analyzing energy consumption across states, sectors, and sources. The project identifies seasonal patterns, renewable vs. non-renewable usage, and regional demand, delivering actionable insights through interactive dashboards.

❖ Dataset Overview:

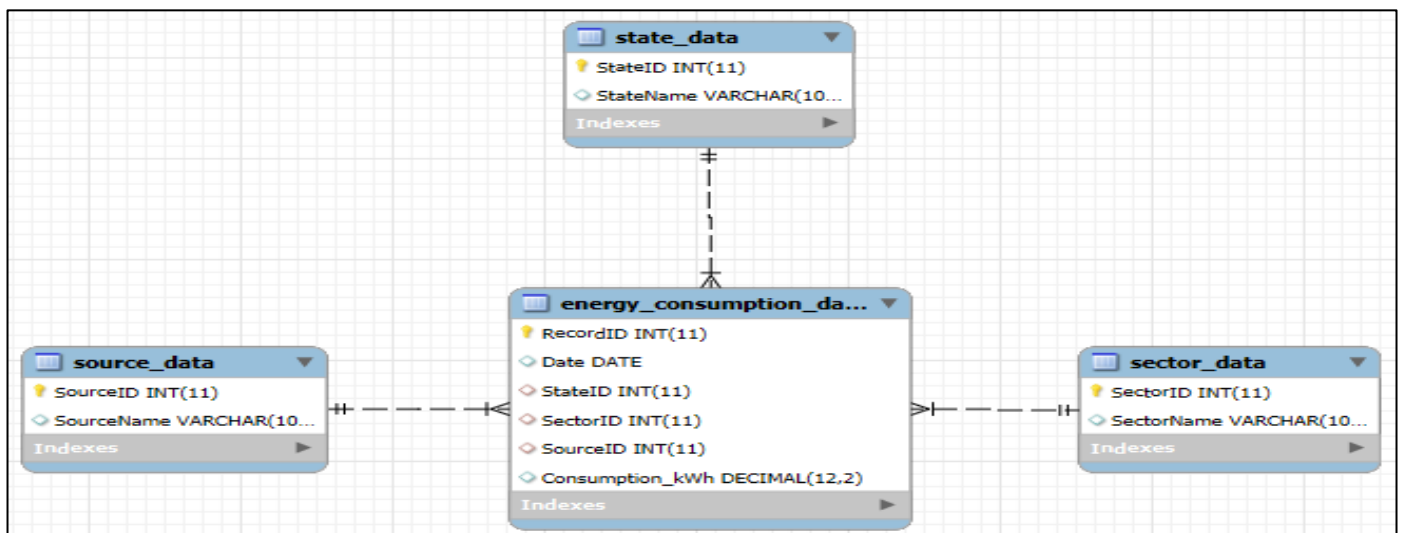
- Time Period: 2020 – 2024 (monthly records)
- Total Records: 1000 rows (200 per year)
- Tables Used:
 - State_Data → Maharashtra, Gujarat, Delhi, Rajasthan
 - Sector_Data → Commercial, Industrial
 - Source_Data → Solar, Wind, Coal, Nuclear
 - Energy_Consumption_Data → Main fact table linking all dimensions

❖ Tools & Technologies:

- Dataset: 4 CSV files (State, Sector, Source, Energy Consumption)
- Database: MySQL Workbench (data import, schema design, SQL queries)
- Visualization: Power BI (data modeling, KPIs, interactive dashboards)

❖ Database Schema / ER Diagram:

- The project follows a snowflake schema design.
- Fact Table: Energy_Consumption_Data (stores detailed monthly consumption records).
- Dimension Tables:
 - State_Data (Maharashtra, Gujarat, Delhi, Rajasthan)
 - Sector_Data (Commercial, Industrial)
 - Source_Data (Solar, Wind, Coal, Nuclear)
- Relationships: The dimension tables are linked to the fact table using foreign keys (StateID, SectorID, SourceID).
- This structure ensures normalized data, avoids redundancy and supports efficient queries in MySQL and Power BI.



❖ SQL Implementation:

1) Table Creation:

➤ Created four tables in MySQL Workbench:

- State_Data
- Sector_Data
- Source_Data
- Energy_Consumption_Data

- Defined Primary Keys in each dimension table. Added Foreign Keys in Energy_Consumption_Data to link with State_Data, Sector_Data, and Source_Data.
- Used AUTO_INCREMENT for ID fields in dimension tables for unique identification.
- Ensured proper data types (INT, VARCHAR, DATE, DECIMAL) for consistency.

```
4 • CREATE TABLE State_Data (  
5     StateID INT PRIMARY KEY,  
6     StateName VARCHAR(100));  
7  
8 • CREATE TABLE Sector_Data (  
9     SectorID INT PRIMARY KEY,  
10    SectorName VARCHAR(100));  
11  
12 • CREATE TABLE Source_Data (  
13     SourceID INT PRIMARY KEY,  
14     SourceName VARCHAR(100));  
15  
16 • CREATE TABLE Energy_Consumption_Data (  
17     RecordID INT PRIMARY KEY,  
18     Date DATE,  
19     StateID INT,  
20     SectorID INT,  
21     SourceID INT,  
22     Consumption_kwh DECIMAL(12,2),  
23     FOREIGN KEY (StateID) REFERENCES State_Data(StateID),  
24     FOREIGN KEY (SectorID) REFERENCES Sector_Data(SectorID),  
25     FOREIGN KEY (SourceID) REFERENCES Source_Data(SourceID));
```

2) Data Insertion (CSV Import / Queries):

- The dataset was prepared as four CSV files: State_Data, Sector_Data, Source_Data, and Energy_Consumption_Data.
- Data was imported into MySQL Workbench using the Table Data Import Wizard, ensuring correct mapping of columns to table fields.
- Each table was successfully populated with records:
 - State_Data – 4 rows
 - Sector_Data – 2 rows
 - Source_Data – 4 rows
 - Energy_Consumption_Data – 1000 rows (2020–2024, 200 per year)
- After import, sample checks confirmed that all rows were inserted correctly.

MySQL Workbench

PowerBI_Project - Warning - ... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

hardik_main

pbi

Tables

energy_consumption_data

sector_data

source_data

state_data

Views

Stored Pro

Functions

phpmyadmin

Administration

Sch

Information

Table: state_data

Columns: StateID, StateName

Select Rows - Limit 5000

Table Inspector

Copy to Clipboard

Table Data Export Wizard

Table Data Import Wizard

Send to SQL Editor

Create Table...

Create Table Like...

Alter Table...

Table Maintenance...

Drop Table...

Truncate Table...

Search Table Data...

Refresh All

Energy_Analysis*

Limit to 5000 rows

```

8 CREATE TABLE Sector_Data (
9     SectorID INT PRIMARY KEY,
10    SectorName VARCHAR(100));
11
12 CREATE TABLE Source_Data (
13     SourceID INT PRIMARY KEY,
14    SourceName VARCHAR(100));
15
16 CREATE TABLE Energy_Consumption_Data (
17     RecordID INT PRIMARY KEY,
18     Date DATE,
19     StateID INT,
20     SectorID INT,
21     SourceID INT,
22     Consumption_kWh DECIMAL(12,2),
23     FOREIGN KEY (StateID) REFERENCES State_Data(StateID),
24     FOREIGN KEY (SectorID) REFERENCES Sector_Data(SectorID),
25     FOREIGN KEY (SourceID) REFERENCES Source_Data(SourceID));
26
27 SELECT * FROM State_Data;
28 SELECT * FROM Sector_Data;
29 SELECT * FROM Source_Data;
30 SELECT * FROM Energy_Consumption_Data;

```

Object Info Session

Ready

```

1039 SELECT * FROM State_Data;
1040 SELECT * FROM Sector_Data;
1041 SELECT * FROM Source_Data;
1042 SELECT * FROM Energy_Consumption_Data;

```

Result Grid

Filter Rows:

Edit:

Export/Imp

	RecordID	Date	StateID	SectorID	SourceID	Consumption_kWh
▶	1	2020-05-16	3	2	1	4108.62
	2	2020-10-01	1	2	1	1042.87
	3	2020-03-30	4	2	1	3569.05
	4	2020-12-19	2	2	1	2928.92
	5	2020-11-11	2	1	1	3998.82
	6	2020-06-20	1	2	1	1656.57
	7	2020-11-14	2	1	1	1594.44
	8	2020-06-21	3	1	1	3933.24
	9	2020-08-10	1	2	1	2683.05
	10	2020-01-13	2	1	1	3673.99
	11	2020-12-29	3	1	1	4706.59
	12	2020-10-03	1	2	1	4646.03
	13	2020-08-08	2	2	1	4271.60
	14	2020-01-17	2	2	1	3635.89
	15	2020-10-02	1	1	1	2880.87
	16	2020-10-22	3	2	1	3427.50
	17	2020-03-25	2	1	1	4932.80
	18	2020-01-30	4	2	1	1065.32

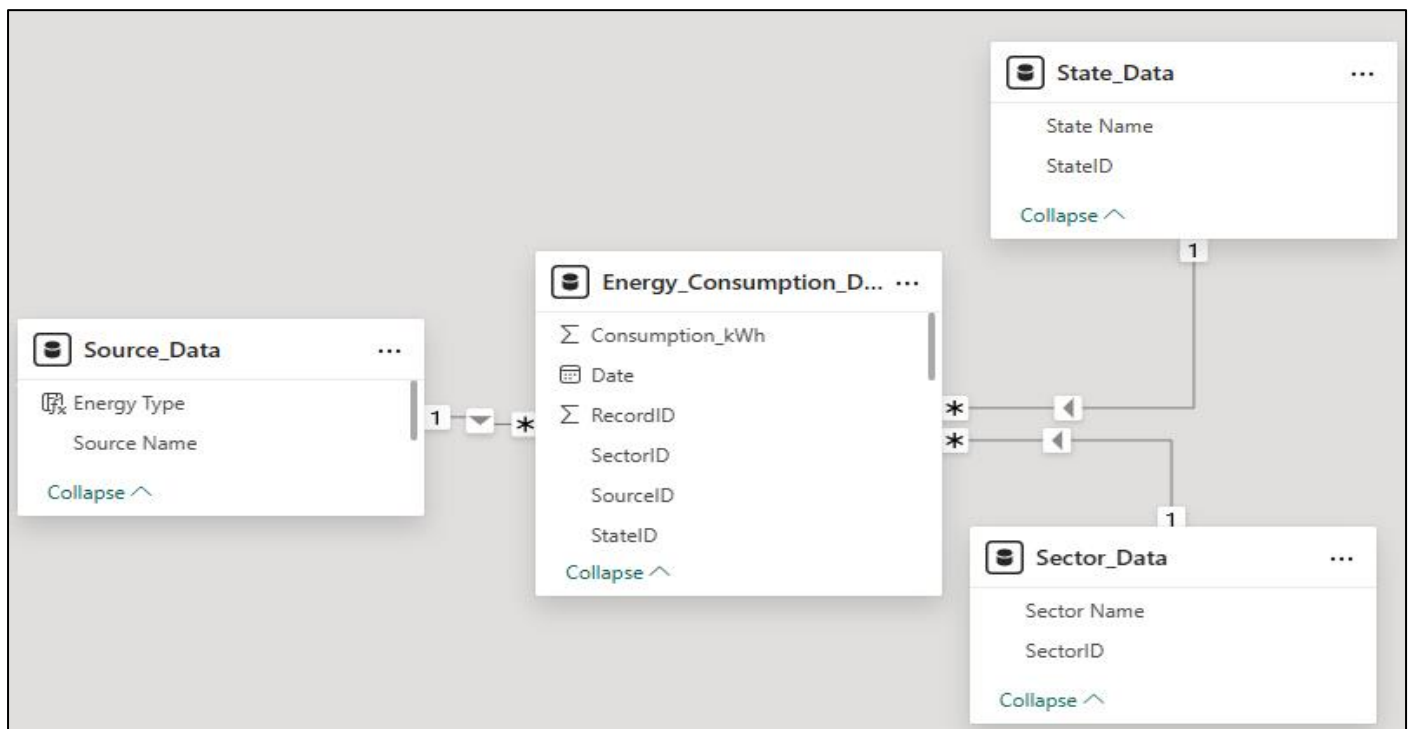
❖ Power BI Data Modeling:

➤ Import from MySQL:

- Connected Power BI to MySQL Workbench database PBI.
- Imported four tables: State_Data, Sector_Data, Source_Data, and Energy_Consumption_Data.
- Ensured all fields were correctly recognized (Date as Date, IDs as Integer).

➤ Relationship Building (Snowflake Schema):

- Established relationships:
 - Energy_Consumption_Data.StateID → State_Data.StateID
 - Energy_Consumption_Data.SectorID → Sector_Data.SectorID
 - Energy_Consumption_Data.SourceID → Source_Data.SourceID
- This snowflake schema allowed easy filtering and aggregation in Power BI.



❖ Report Pages:

➤ Page 1: Time Series & Seasonal Trends:

- This page focuses on temporal energy consumption patterns. Multiple visuals track monthly and yearly trends, average vs total consumption and seasonal variations. KPIs highlight average monthly usage, peak consumption month and total energy consumption. These insights help identify demand fluctuations and peak periods for better planning.

➤ Page 2: Sector-wise Analysis:

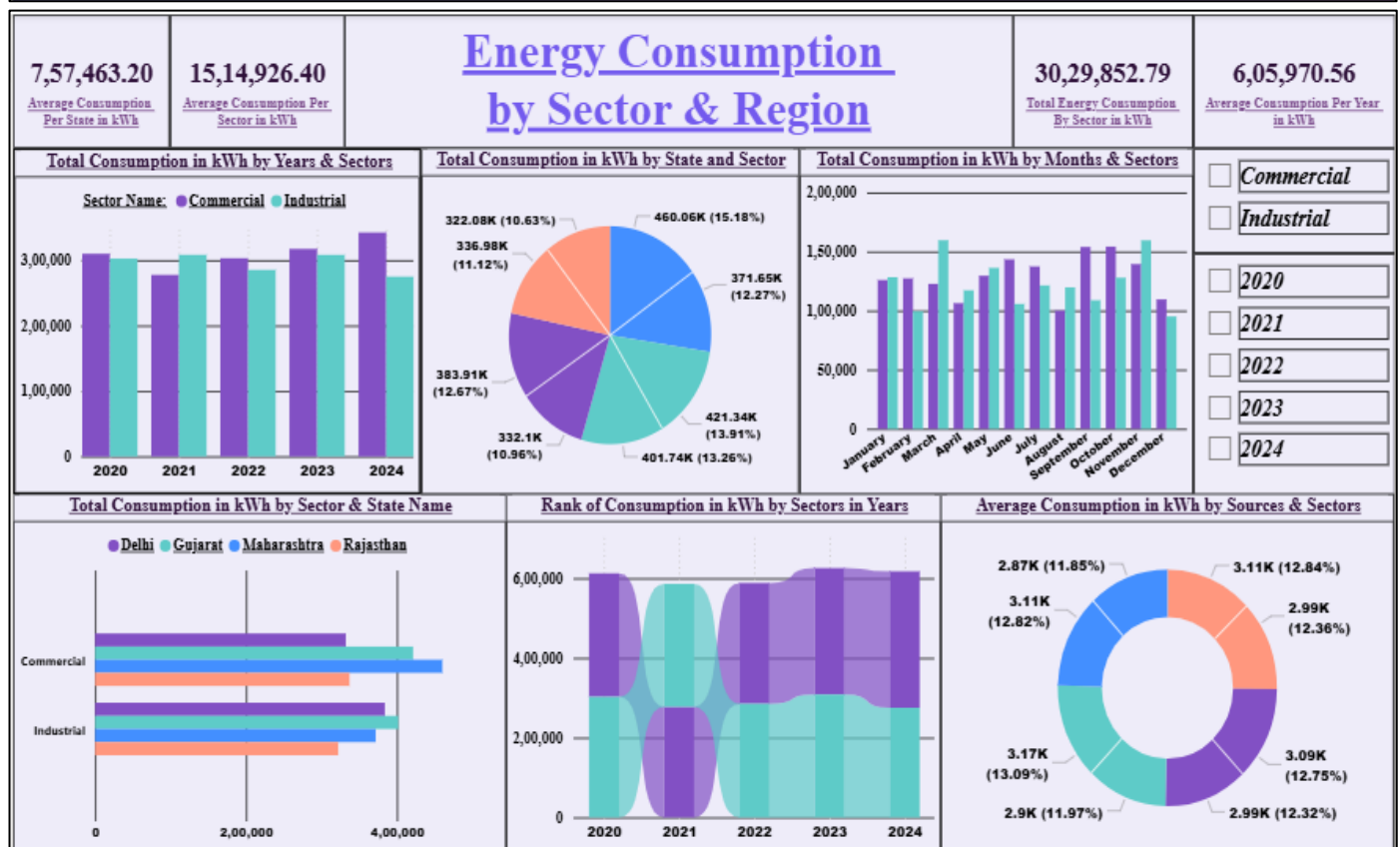
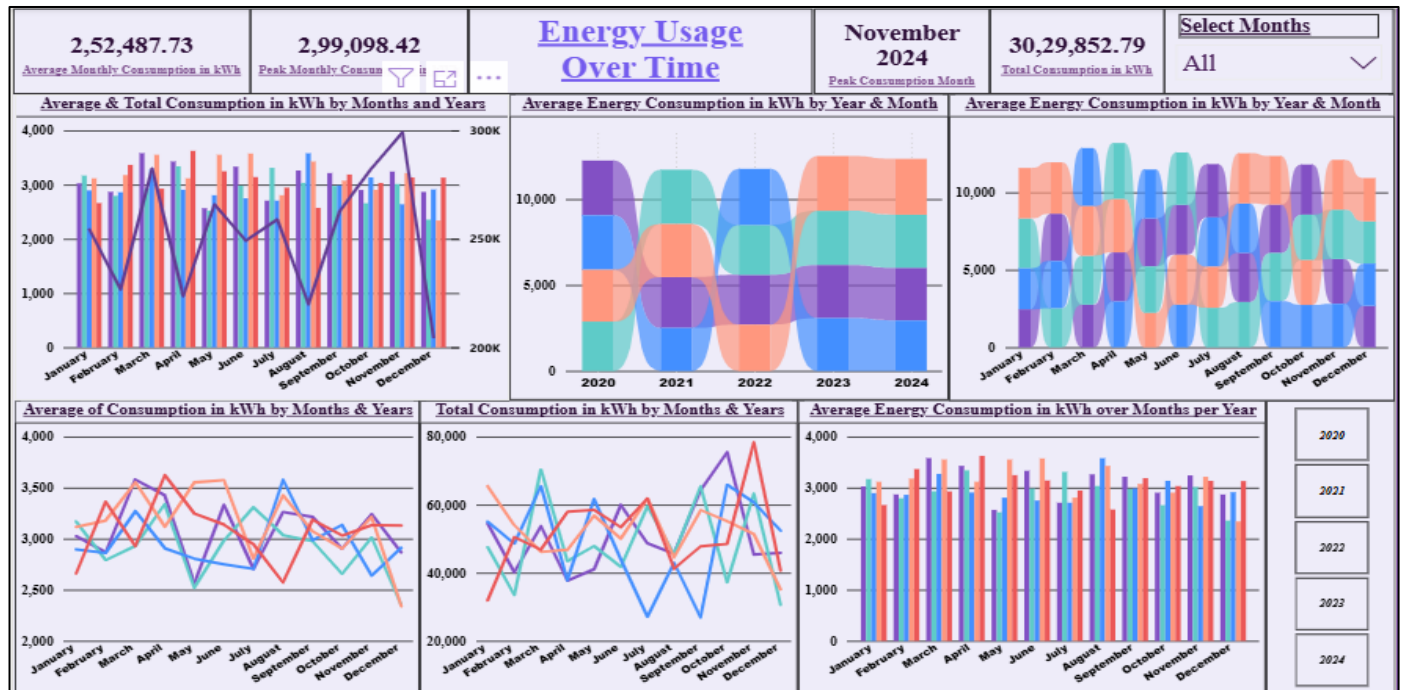
- This page analyzes energy consumption across Commercial and Industrial sectors. Visuals include comparisons by year, state, and month, supported by sectoral KPIs. Donut and bar charts highlight proportional shares and state-level differences. The analysis identifies the most energy-intensive sectors and reveals growth trends in different regions.

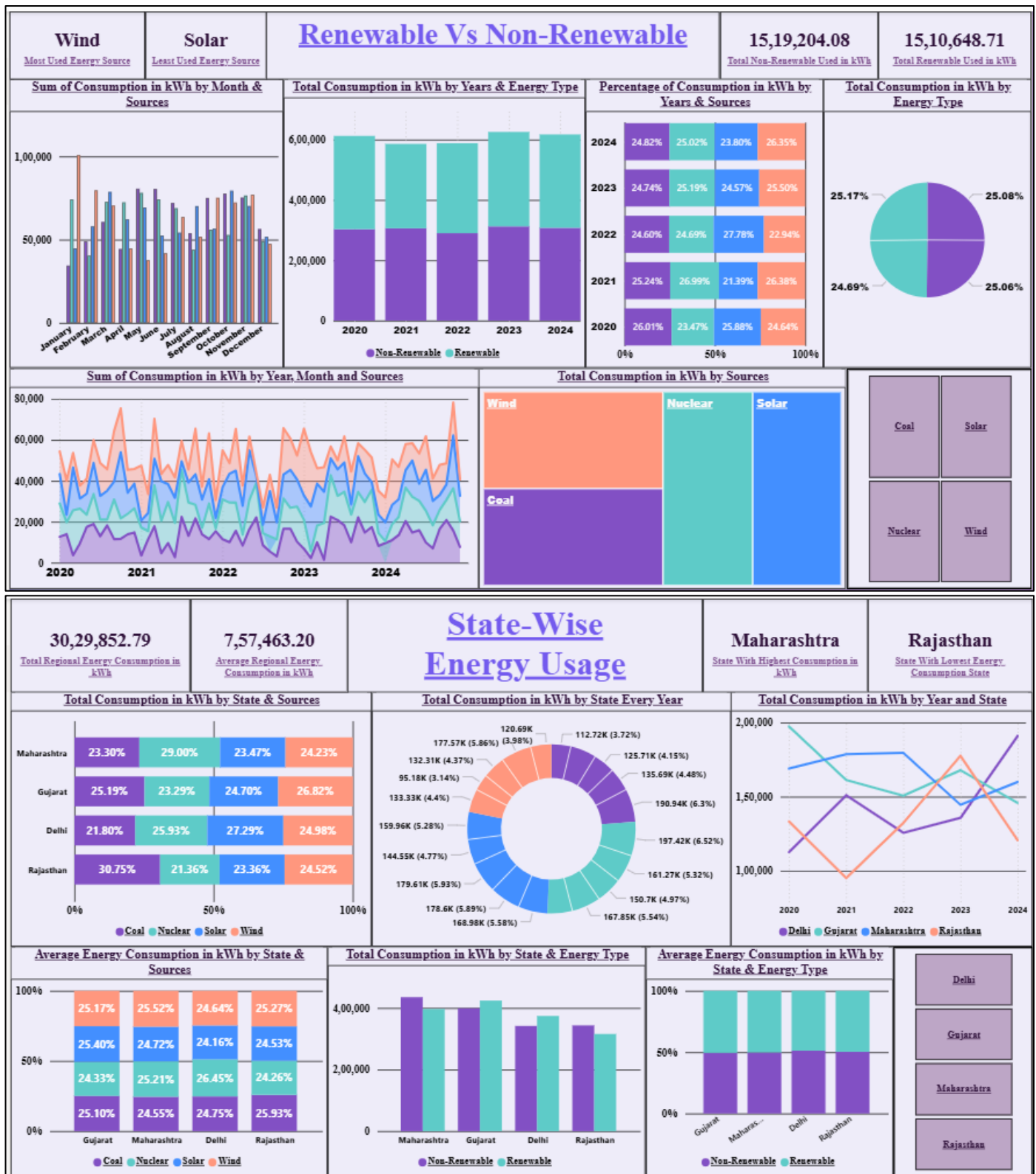
➤ Page 3: Source-wise Analysis:

- This page compares renewable sources (Solar, Wind) and non-renewable sources (Coal, Nuclear). Stacked charts, line graphs, treemaps and KPIs display overall usage, share percentages, and year-wise adoption. The report highlights the most and least used energy sources and tracks renewable adoption progress over five years.

➤ Page 4: Regional Consumption:

- This page examines state-level energy demand across Maharashtra, Gujarat, Delhi, and Rajasthan. Visuals include pie charts, line graphs, and bar comparisons of sources and energy types per region. KPIs display the highest and lowest consuming states. The analysis supports location-specific planning and highlights regional disparities in renewable vs non-renewable usage.





Average Energy Consumption in kWh by State & Sources

Total Consumption in kWh by State & Energy Type

Average Energy Consumption in kWh by State & Energy Type

Delhi

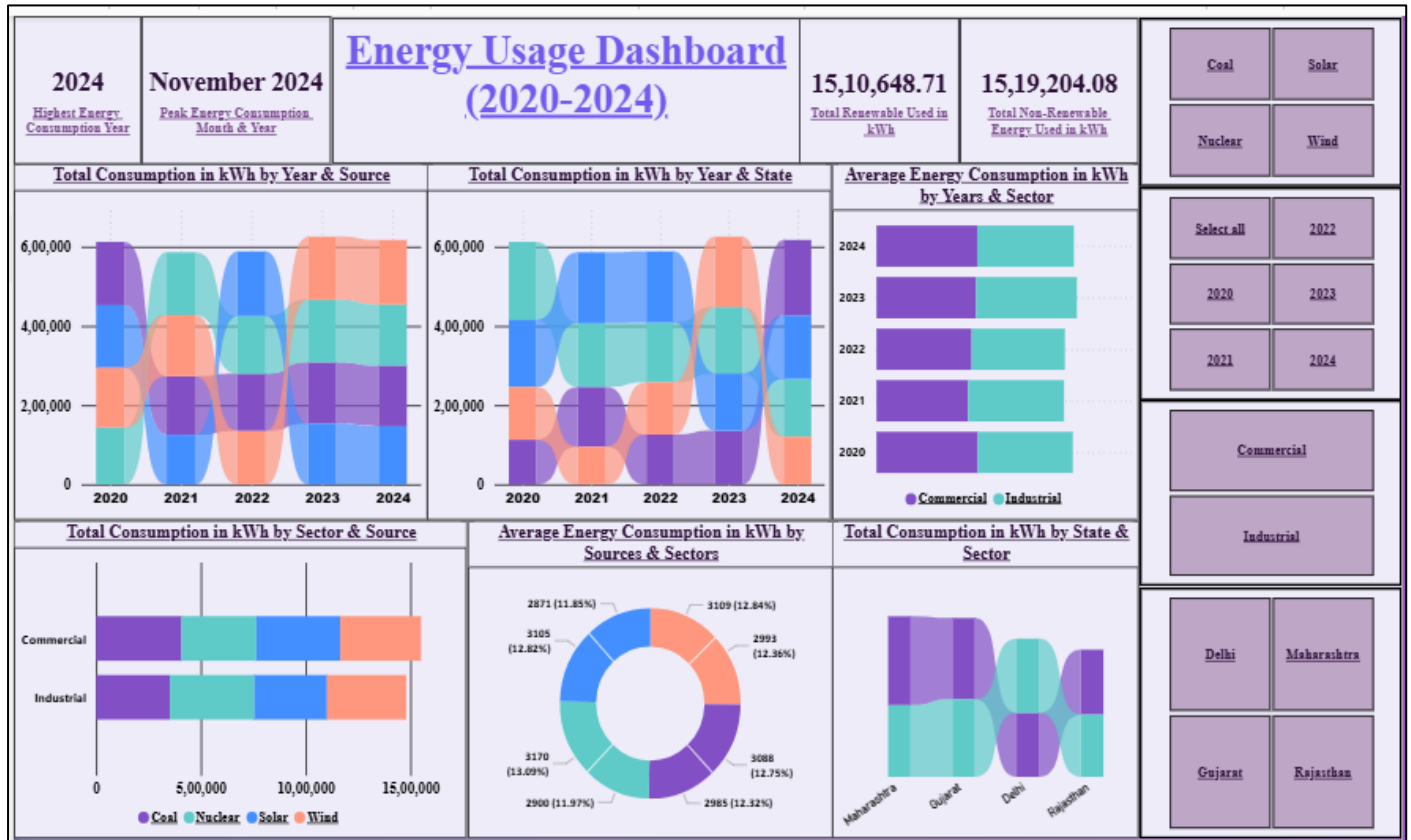
Gujarat

Maharashtra

Rajasthan

❖ Summary Dashboard (Page 5):

- The summary dashboard consolidates insights from all report pages into a single interactive view for decision-making. It highlights key KPIs including highest consumption year, peak month and renewable vs non-renewable usage. Visuals such as stacked area charts, bar charts, donut charts and treemaps provide a clear breakdown of trends by year, sector, source and state. Interactive slicers further enable dynamic filtering, making the dashboard flexible and user-friendly. From this analysis, 2024 emerged as the highest energy usage year, with Industrial sector dominance and Maharashtra leading among states.



❖ Key Insights / Analysis Performed:

- Peak demand observed during summer months across all states.
- Industrial sector consumed the highest share of energy.
- Renewable energy (Solar & Wind) adoption steadily increased year over year.
- Maharashtra showed consistently higher consumption compared to other states.

❖ Outcome:

- Delivered a scalable energy analytics pipeline integrating CSV → MySQL → Power BI.
- Designed four detailed report pages and one summary dashboard for comprehensive energy monitoring.
- Enabled KPI tracking for peak demand, renewable adoption, sectoral consumption and regional trends.
- Provided interactive, user-friendly dashboards supporting data-driven decision-making.

❖ Conclusion:

- This project successfully demonstrated the end-to-end workflow of data engineering and analytics, starting from raw CSV ingestion to interactive dashboarding.
- A normalized schema was designed in MySQL to ensure data consistency, scalability, and integrity, forming the backbone of analysis.
- By connecting MySQL to Power BI, the project enabled dynamic reporting, KPI tracking and trend visualization across sources, sectors, and regions.
- Insights revealed clear patterns such as peak seasonal demand, industrial sector dominance and growing renewable adoption.
- The solution highlights how integrating SQL data modeling with BI tools can transform raw data into actionable intelligence.
- The pipeline is flexible and can be further scaled to real-time monitoring, larger datasets or advanced predictive analytics, supporting both operational decision-making and long-term policy planning.